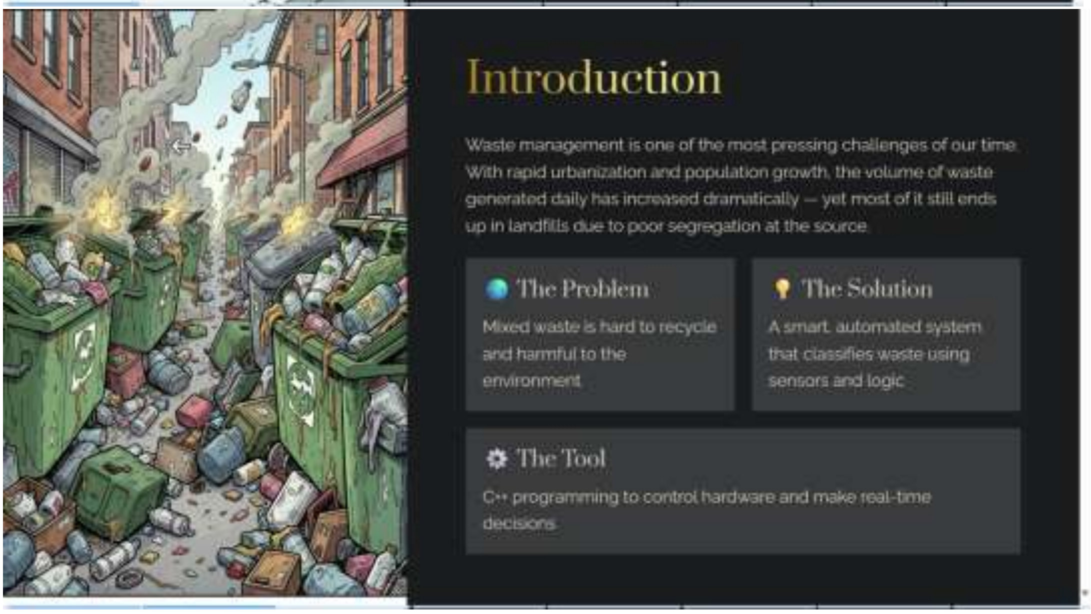
TECHNICAL PRESENTATION

-AFEIA JUSTIN JOSE LITTLE BRIGHT (1925244494)



Problem Statement

- Most people do not sort waste correctly due to lack of awareness or convenience
- Manual sorting is slow, error-prone, and labor-intensive
- Mixed waste contaminates recyclable materials, making them unusable
- Improper disposal leads to soil and water pollution
- Existing smart systems are expensive and not widely adopted

Objectives

This project aims to design a smart waste segregation system using C++ that addresses real-world waste management challenges

1

Automate Classification

Detect and sort waste into categories — biodegradable, recyclable, and non-recyclable — without human intervention

2

Simulate Sensor Logic

Use C++ to model sensor input and decision-making logic for waste type identification

3

Reduce Contamination

Ensure clean separation of waste streams to improve recycling efficiency

4

Provide Feedback

Display real-time status and alerts to users through a simple interface

How the System Works

Waste Detected

Waste Analyzed

Bin Selected

Status Updated

The system uses a combination of sensors (like IR or capacitive sensors) to detect the type of waste. The C++ program reads sensor data, applies classification rules, and activates the appropriate bin mechanism — all in real time.

System Architecture

Input Layer — Sensors

Sensors detect physical properties of waste (density, material type) and send signals to the microcontroller

Processing Layer — C++ Logic

The C++ program processes sensor input, runs classification algorithms, and determines the waste category

Output Layer — Actuators

Servo motors or solenoids direct waste into the correct bin based on the program's decision

Advantages

Environmentally Friendly

Reduces landfill waste and promotes proper recycling

Fast & Accurate

Automated sorting is faster and more consistent than manual methods

Cost-Effective

Reduces long-term labor costs for waste management facilities

Educational Value

Hands-on project combining programming, electronics, and environmental science

Future Scope & Conclusion

Future Enhancements

- Integrate AI/ML for more accurate waste identification using cameras
- Add IoT connectivity for remote monitoring and data analytics
- Deploy in public spaces, schools, and smart cities

✅ Conclusion

This project demonstrates how **C++ programming** can be applied to solve a real-world environmental problem. By automating waste segregation, we can significantly improve recycling rates, reduce pollution, and move toward smarter, more sustainable cities.

Smart technology + environmental awareness = a cleaner future for everyone